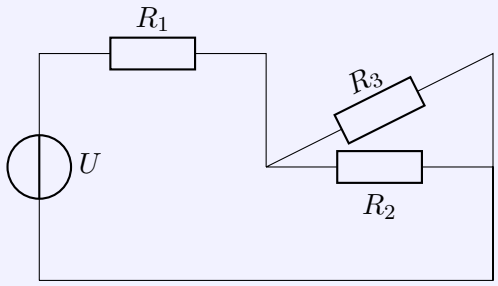
Exercise Sheet #3

DC 2

Exercise 1: Network Analysis 1

Given the circuit with $R_1 = 6\,\Omega$, $R_2 = 12\,\Omega$, $R_3 = 4\,\Omega$ and a voltage source of $24\,\mathrm{V}$:

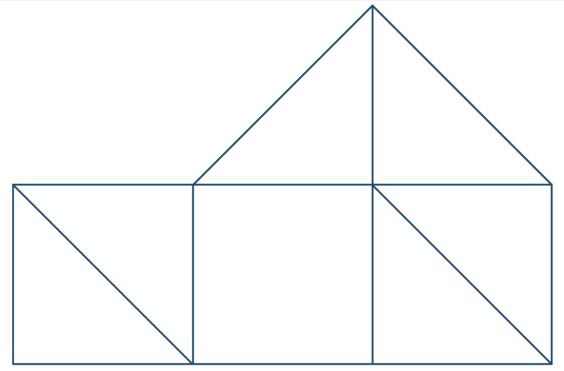


Determine:

1. the total source current,
2. the branch currents through R_2 and R_3,
3. the power dissipated in R_3.

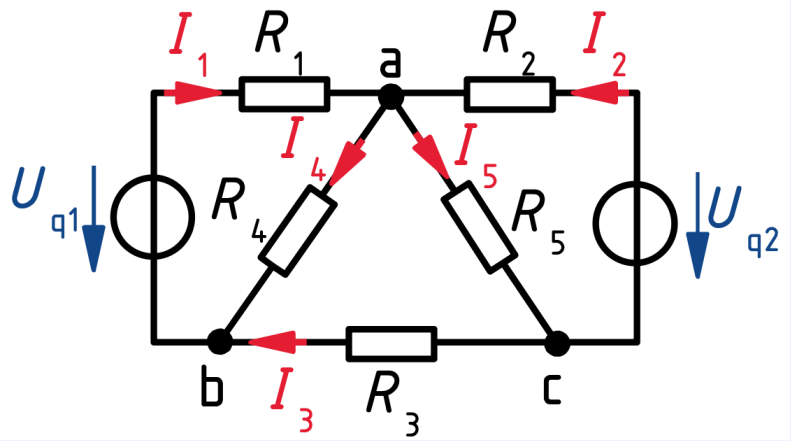
Exercise 2: Topology

Find the number of nodes, branches, and loops.



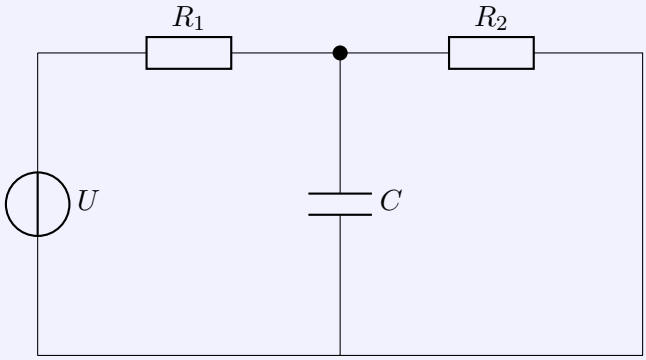
Exercise 3: Network Analysis 2

The following network contains the resistors $R_1 = 1\,\Omega$, $R_2 = 2\,\Omega$, $R_3 = 5\,\Omega$, $R_4 = 25\,\Omega$, and $R_5 = 40\,\Omega$, as well as the source voltages $U_{q1} = 16.2\,\mathrm{V}$ and $U_{q2} = 11.4\,\mathrm{V}$. All branch currents are to be determined.



Exercise 4: Nodal Analysis

Given the DC network with $U = 20\,\mathrm{V}$, $R_1 = 5\,\Omega$, $R_2 = 10\,\Omega$, $C = 100\,\mu\mathrm{F}$:

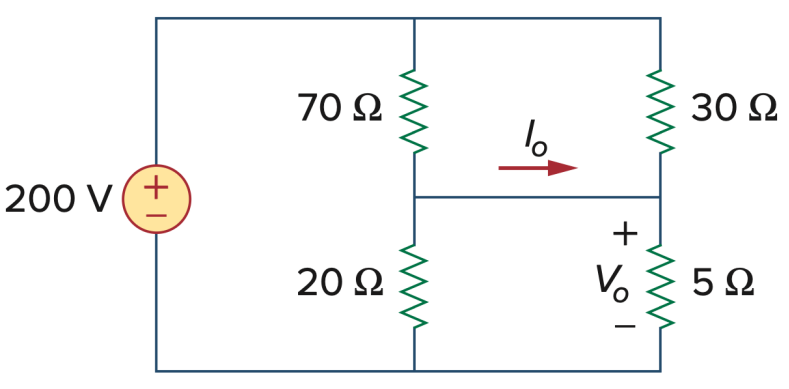


The lower conductor is chosen as reference potential (ground). The steady-state condition ($t \to \infty$) is assumed.
Determine:

1. the node voltage at the marked node,
2. the voltage across the capacitor,
3. the current through the capacitor,
4. the energy stored in the capacitor.

Exercise 5: Node Connection

Calculate I_o and V_o.

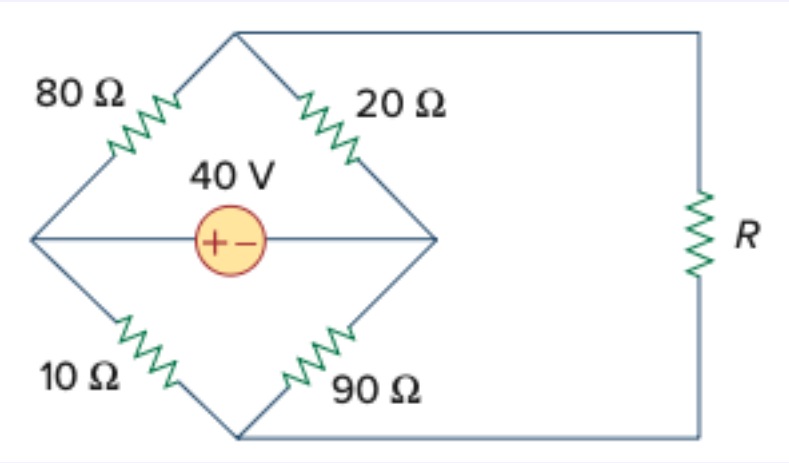


Exercise 6: Thévenin and Power Matching

The variable resistor R is adjusted until it absorbs the maximum power from the circuit.

a) Calculate the value of R for maximum power.

b) Determine the maximum power absorbed by R.



Exercise 7: Efficiency

For a real voltage source with an internal resistance and a connected load, the efficiencies given below apply. For each case, calculate what fraction of the maximum possible power is absorbed by the load.

a) $\eta = 75\%$

b) $\eta = 90\%$

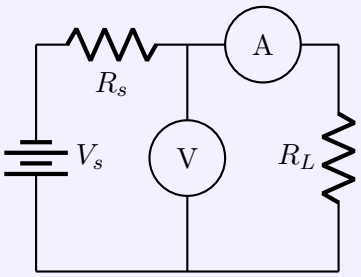
Exercise 8: Power Measurement

Calculate the effect of voltmeter and ammeter placement on the measured power in a DC resistive circuit with the parameters and configuratons below. Compare the measured power to the true power dissipated in the load resistor for both configurations and for the two different load resistances.

- Source voltage: $V_s = 12\,\text{V}$

- Source internal resistance: $R_s = 1\,\Omega$
- Voltmeter internal resistance: $R_v = 1000\,\Omega$
- Ammeter internal resistance: $R_a = 0.1\,\Omega$
- Load resistance (two cases):
  - Low load resistance: $R_L = 1\,\Omega$
  - High load resistance: $R_L = 100\,\Omega$

1. Configuration A (Current-corrected):
   Voltmeter connected across the source, ammeter in series with the load.



2. Configuration B (Voltage-corrected):
   Voltmeter connected across the load, ammeter in series with the source.